STOCK MARKET ANALYSIS

PRELIMNARY REPORT

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ADVISORY AND LIMITATION OF LIABILITY

This document presents a preliminary investigation into stock market analysis using signal processing and machine learning techniques. The results and interpretations are exploratory and derived from third-party datasets that may contain noise, inconsistencies, or sampling limitations. While standard analytical methods have been applied, factors such as data quality, incomplete feature representation (including absence of sentiment inputs), and inherent market volatility may influence the observed outcomes.

Additionally, the implementation of models and data processing pipelines may be subject to unintended errors or limitations typical of early-stage research. Accordingly, the findings should be interpreted as indicative insights rather than definitive conclusions and are not intended as financial advice or investment recommendations. This work serves as a foundation for further refinement, and future iterations may revise or replace aspects of the methodology and results.

– Team Gradient Descent into Madness

SOURCES

It is advised to read github repo : https://github.com/SupreethSuper/ML_proj_imp_on_FPGA.git

This repo contains the codes for the project.

The dataset has been sourced through Kaggle.

PROPOSAL

This project aims to provide a AI model that can be trained to generate an advise weather to hold the stock or sell the stock for the next 2 – 4 months. The project achieves this by feeding an AI algorithm real data of one of the S&P 500 companies, and sentimental news that might hurt or boost the stock, that the user is interested.

PROGRESS

We have progressed and implemented some simple machine learning algorithms to observe what kind of data are we dealing with. We have also implemented a few filters, such as mode 50 avg filter, fft filter and nquist sampler. By implementing this, the preliminary observations are :

1) The data has enormous amount of noise

2) The data tends the model to cause slight overfit

However, the project has not yet implemented the sentimental data, which might be the main cause for the underperforming model, The model is predicting at best 51% using k means, (k=5) cross validation.

RESULTS

The initial PCA reports suggests that there is lot of noise and jitter in the data. However, it does suggest that, during some essential “financial blackouts/crisis” such as the great depression, COVID 19 pandemics.

There are some events that are unknown to us (The team), where the stocks have dropped significantly. This has been captured by all filters mentioned above. The snapshots of PCA, and the filters implemented are below.

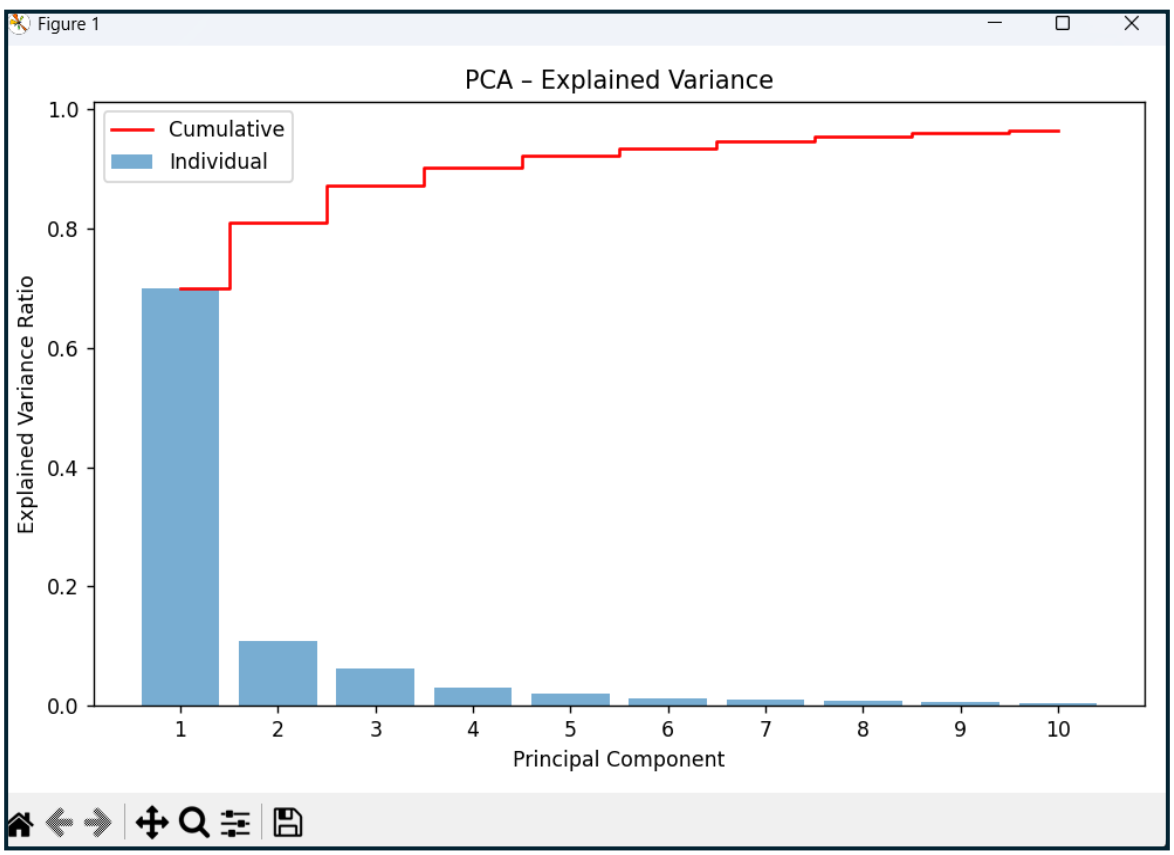

Fig 1 – PCA

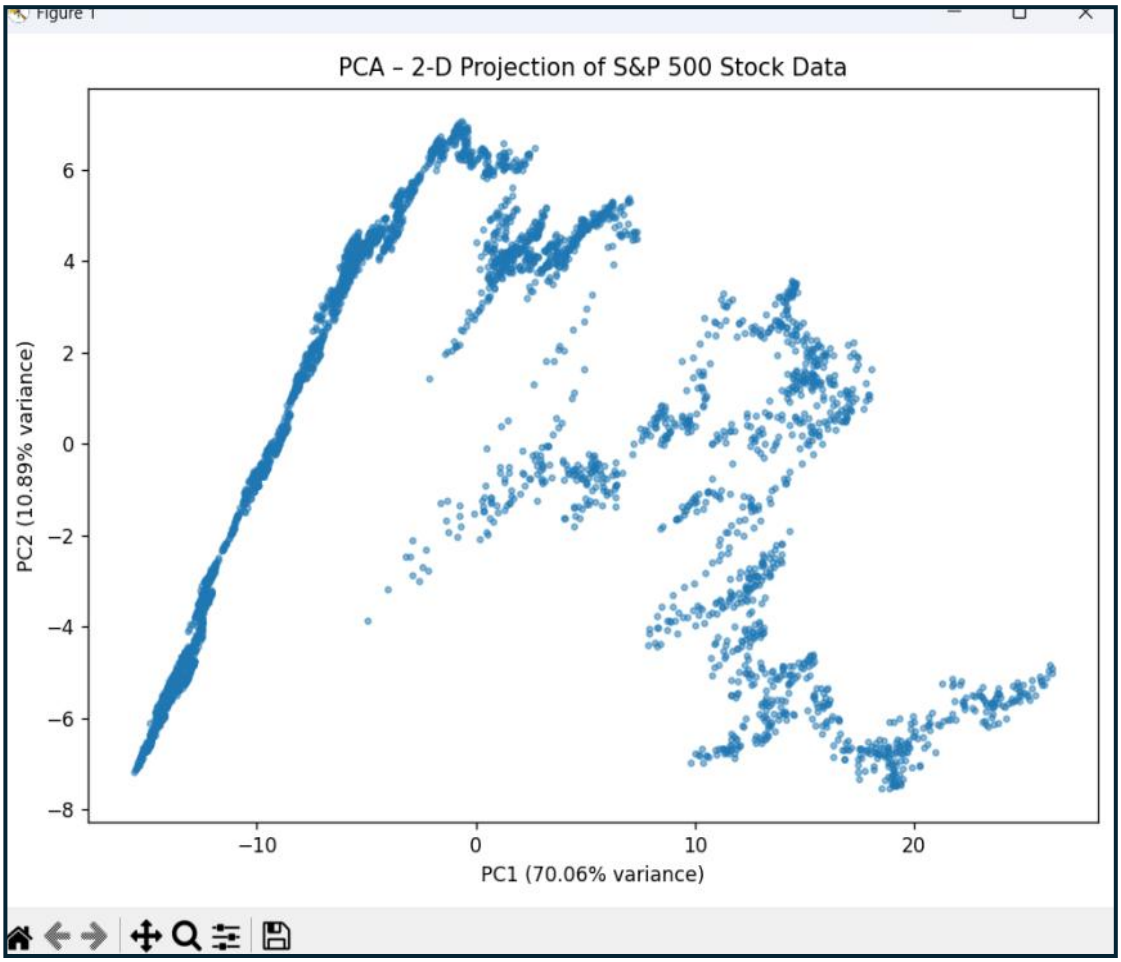

Fig 2 – projection of raw stock data

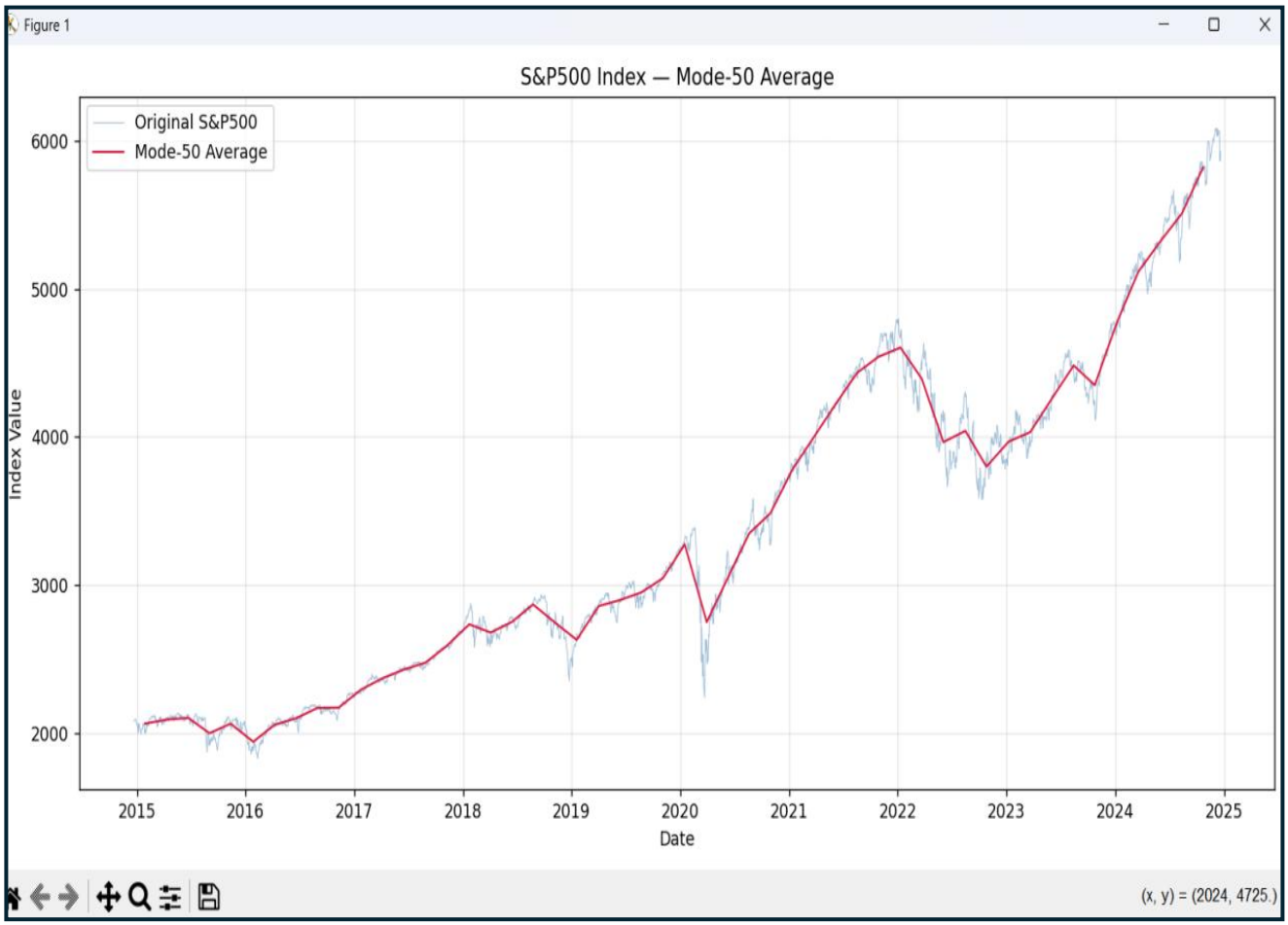

Fig 3 - mode 50 avg filter

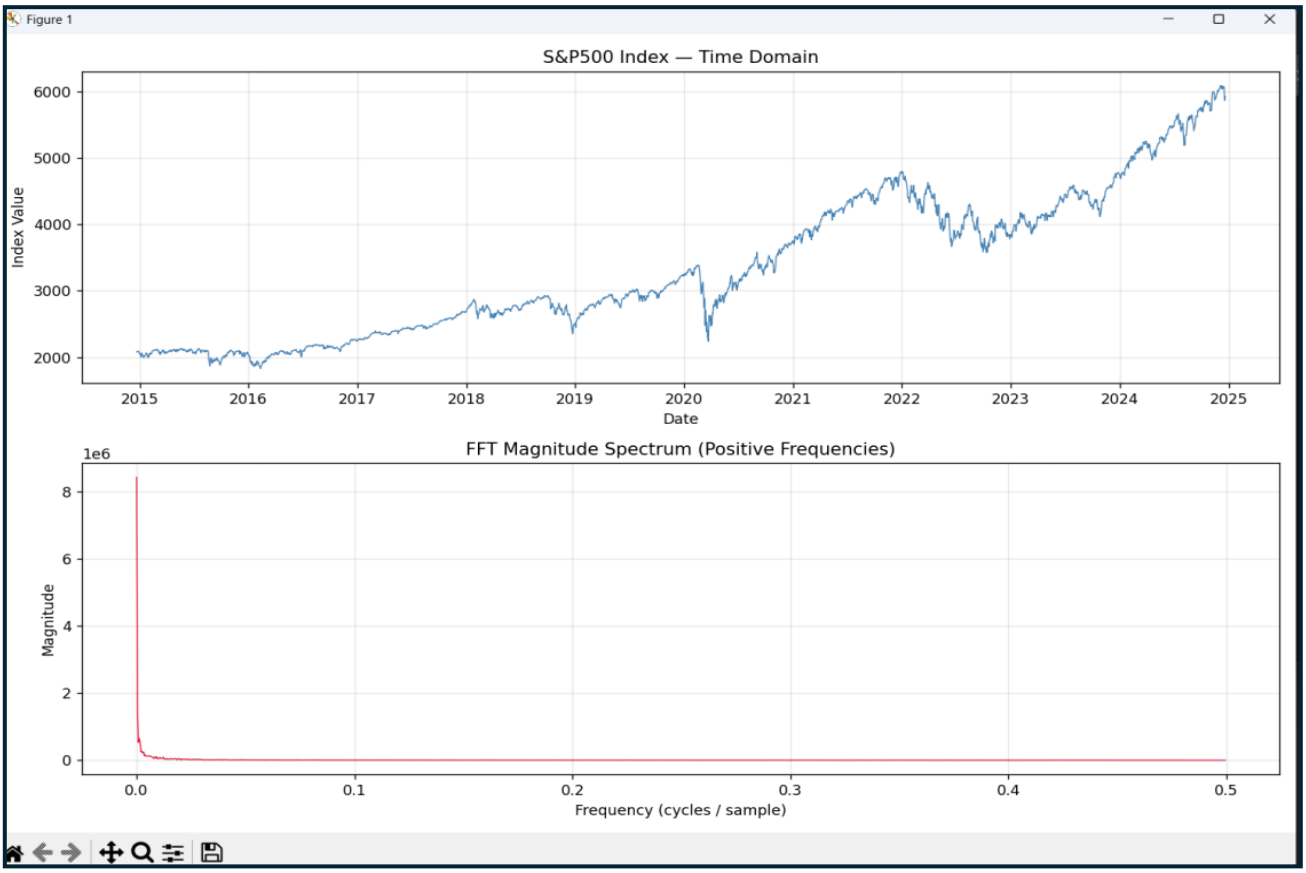

Fig 4 - fft on data

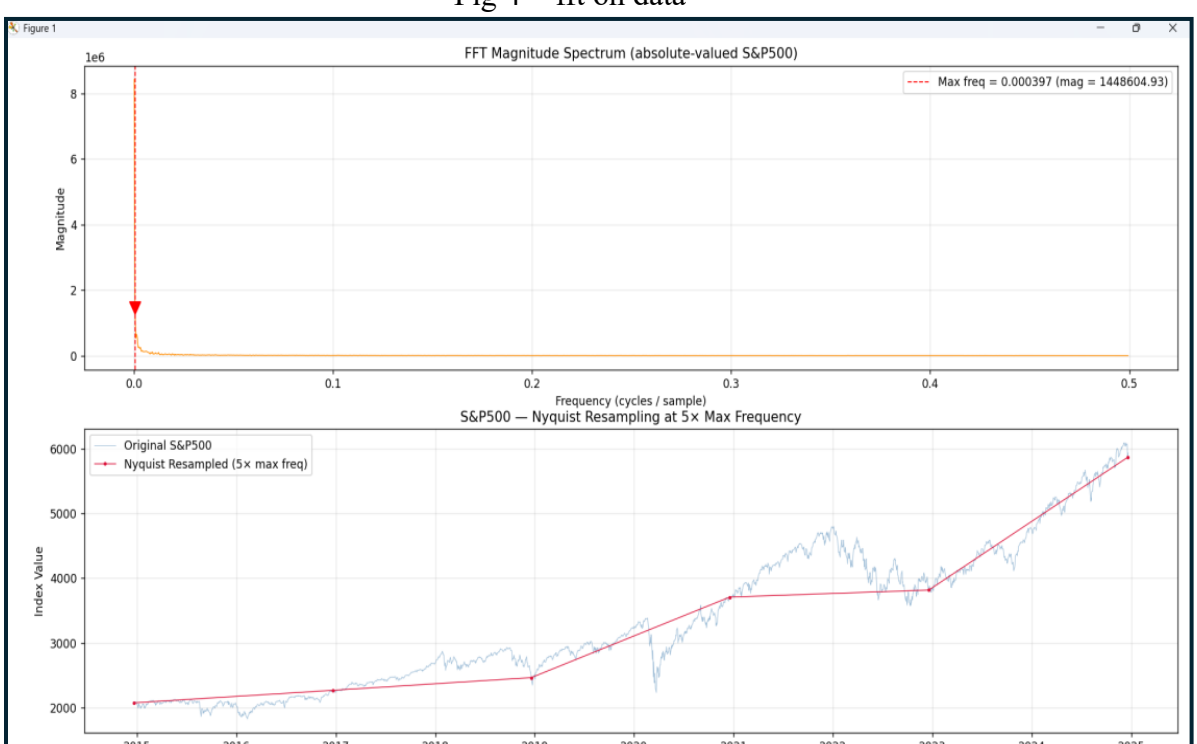

Fig 5 – Nyquist sampler

All the above results are from a filters, with an exception that is PCA, but PCA is more of data optimizer, that scrubs the data, and is mostly used for our case of high variance data.

For datasets like these, Its strongly advised to use unsupervised learning to create a better understanding of the data, and create better discrimination of the data. The labeled data provided by unsupervised learning can be fed to supervised learning, to get a learning outcome. With 1 week of work, the project has completed trials till only one of the unsupervised learning methods – KNN with 5-fold.

The results are published below

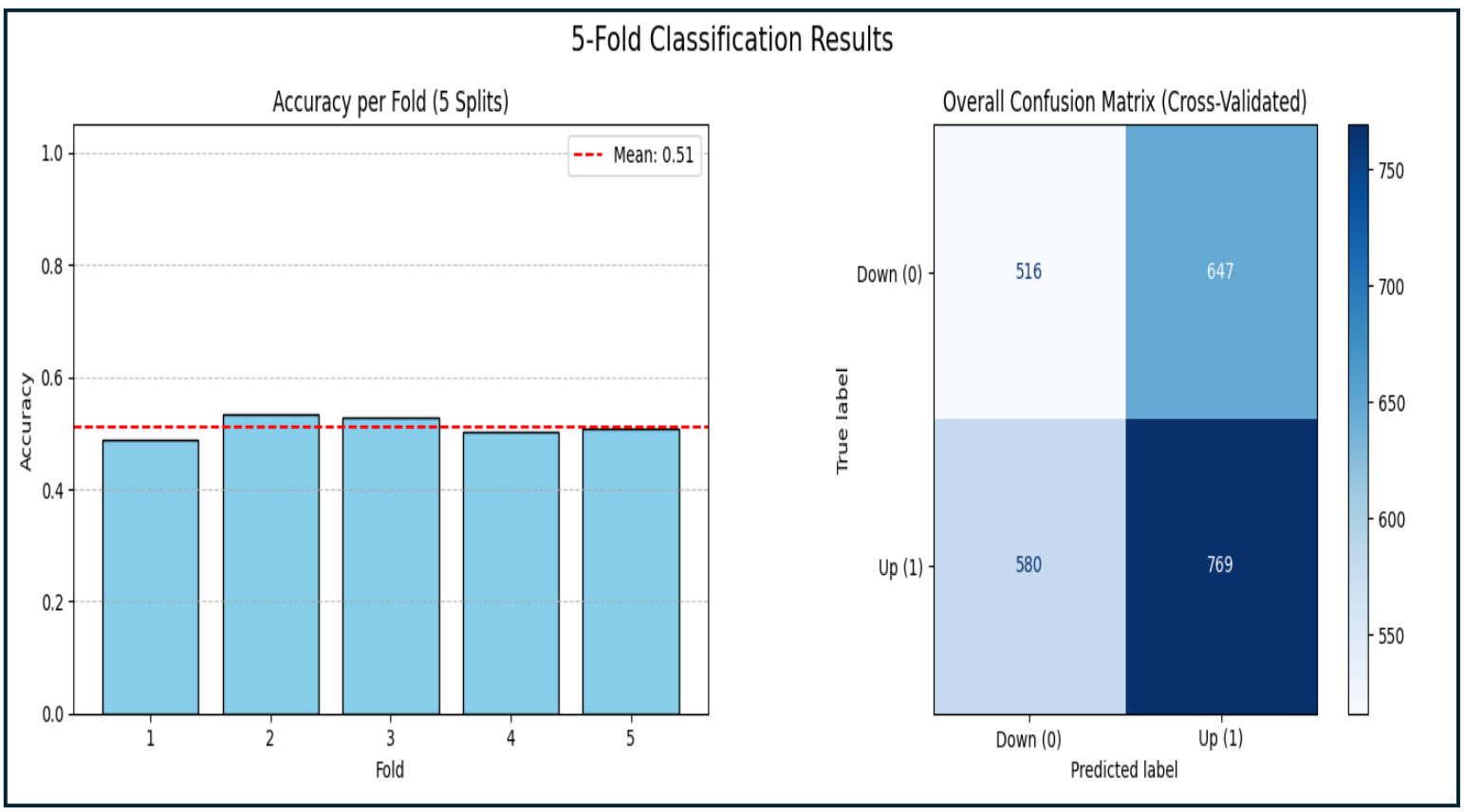

Fig 6 – Confusion matrix of KNN

FURTHER WORK

- ➔ The project aims to find the optimal unsupervised learning algorithm, for data discrimination.
- ➔ The project aims to find a reliable data set for processing sentiments, hereby referred to as “sentimental analysis”, through the rest of the report.
- ➔ The usage of unsupervised learning may show its dark side, as it may not be able to classify such high variance data into perfect labels, this can be a tradeoff for confusion matrix of supervised learning